F 4/2

MIL-C-46946A(MU)
3 December 1973
SUPERSEDING
MIL-C-46946(MU)
25 September 1963

MILITARY SPECIFICATION

CARTRIDGE, IMPULSE, M43, ASSEMBLY

1. SCOPE

1.1 This specification covers the Cartridge, Impulse, M43 for use with propellant actuated devices in aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specification for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Coating, Anodic Conventional For Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MS-28741	- Hose, Assembly, Detachable, End Fitting, Medium Pressure

DRAWINGS

Armament Command

D8593275	- Cartridge, Impluse, M43, Assembly

FSC 1377

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:23Z
Check the source to verify that this is the current version before use.

95

2.1 (Cont'd)

DRAWINGS (Cont'd)

Armaments Command

F8797442	- Box, Packing, Ammunition, for Cartridge, Impulse, M43
F8797443	- Carton, Packing, Ammunition, for Cartridge, Impulse, M43
C9225902	- Carton, Packing, Alternative for Cartridge, Impulse, M43
IEL-8593275	- Inspection Equipment List for Cartridge, Impulse, M43

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.-The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

ANSI-B46.1 - Surface Texture

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N. Y. 10018.)

3. REQUIREMENTS

3.1 General.-Materials, parts, and assemblies, shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.-Protective finish shall be in accordance with Specification MIL-C-60539.

3.3 Vibration.-The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII, without evidence of any disintegration of the primer, igniter or propellant or derangement of components.

3.4 Functional requirements.

3.4.1 The cartridge shall not fail to fire.

3.4.2 Functioning.-The cartridge when assembled in an M2A2 Thruster and fired under load (see 6.1) at -65°F to -70°F and at 160°F to 165°F shall be capable of delivering a stroke of 5.7 inches minimum, and move an approved ejection seat either upward or downward with a maximum final velocity of 12 feet per second (fps) so that the ejection seat will be in a final position for ejection.

Source: https://assist.dla.mil Downloaded: 2026-10-22T07:23Z
Check the source to verify that this is the current version before use.

95

3.4.3 Initiation.-The thruster shall be fired by a source delivering an output equal to that of an M3A1 Initiator at the end of 15 feet of aircraft hose, Size 4, in accordance with MS28741 at 70°F plus/minus 5°F.

3.5 Workmanship.-The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the following:

3.5.1 Metal defects.-All metal parts shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.5.2 Burr.-The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.5.3 Foreign matter.-The part or assembly shall not contain chips, dirt, grease, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services confrom to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.-Reference shall be made to Standard MIL-STD-109 to define quality assurance terms used.

4.1.2 Inspection.-Inspection shall be in accordance with Specification MIL-A-2550.

4.2 First Article Sample.

4.2.1 Initial production sample.-At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 30 cartridges for ballistic tests, 12 for vibration tests (to be selected from ballistics sample) 5 assemblies for sealant tests,and 10 sets of unassembled components for dimensional and visual examination and tests.

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing, shall be the same as used for regular production and shall be obtained from the same source of supply.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:23Z
Check the source to verify that this is the current version before use.

95

4.2.1.1 Examination and Tests.-After inspection and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.-The sample for ballistic tests shall be temperature conditioned and tested to determine compliance of 3.4. The sample for firing shall consist of the following:

TABLE I

Temperature	Vibrated	Not Vibrated
-65°F to -70°F	4	6
65°F to 75°F	4	6
160°F to 165°F	4	6

4.2.1.3 Initial production sample failure.-Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Presentation of lot.-The product shall be presented in accordance with Standard MIL-STD-105.

4.3.1.2 Size of lot.-A lot shall consist of an identifiable group of assembled units and shall include:

a. Components produced by a homogeneous process.

b. Propellant from not more than one lot.

c. Igniters from not more than one lot.

e. Primers from not more than one lot.

4.3.2 Examination.-One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects, using the sampling plans and criteria in

S/S

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:23Z
Check the source to verify that this is the current version before use.

4.3.2 (Cont'd)

Tables II and III. For an individual characteristic, the criteria in Table II shall be used. For total characteristics within a major or minor group, the criteria in Table III shall be used. All nonconforming material shall be rejected. Four power (4X) magnifications may be used in performing visual examination for such defects as burr and foreign matter. ANSI B46.1 shall be used as a basis of comparison for surface roughness determination.

TABLE II

Lot Size	Sample Size	Acceptance Numbers Major Accept	Major Reject	Minor Accept	Minor Reject
1 -- 300	100% Examination	0	0	0	0
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

TABLE III

Total Allowable Defects Per Sample Size

Sample Sizes for Major Defects - - -	110	150	225	300
Number of Characteristics in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample Sizes for Minor Defects - - -	**110**	**150**	**225**	**300**
Number of Characteristics in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:23Z
Check the source to verify that this is the current version before use.

4.3.2.1 Classification of defects.

4.3.2.1.1 Cartridge, Impulse, M43 Assembly, Drawing D8593275.

Defect	Method of Inspection
CRITICAL	
1. Any part missing	X-ray/Visual 1/
2. Improper assembly	X-ray/Visual 1/
3. Mixed ammunition	Visual
4. Metal defective	Visual
5. Crimp missing or incorrect	Visual
6. Marking missing, incorrect or illegible	Visual
MAJOR	
101. Cartridge profile failure, requiring more than 200 pounds deadweight to insert in concentricity receiver gage	Gage
102. Diameter of body, max.	Gage
103. Diameter of head, max.	Gage
104. Length to datum diameter, max.	Gage
105. Dent, failure to chamber in concentricity receiver gage with 20 pounds deadweight	Gage/Visual
106. "O" Ring split, cracked or cut	Visual
107. Protective finish inadequate or defective	Visual
108. Foreign matter	Visual
109. Burr	Visual
MINOR	
201. Total length	Gage
202. Dent, which will chamber in concentricity receiver gage with 20 pounds deadweight, maximum	Gage/Visual
203. Draw scratch	Visual
204. Fold, wrinkle, buckle or bulge	Visual
205. Varnish inadequate or missing	Visual
206. Surface finish improper	Visual

1/ The assembled cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negative, properly identified, shall be forwarded to the procuring activity.

4.3.1.2 Primer, Percussion, Drawing D10551686, prior to assembly into a cartridge.

CRITICAL	
1. Improper assembly	X-ray 1/

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:23Z
Check the source to verify that this is the current version before use.

S/S

4.3.2.1.2 (Cont'd)

	Defect	Method of Inspection
2.	Any part missing	X-ray 1/
3.	Mixed primers	Visual
4.	Primer pellet broken, cracked or dislodged	Visual
5.	Primer foil discolored	Visual
6.	Metal defective	Visual

MAJOR

101.	Burr	Visual
102.	Foreign matter	Visual

MINOR

None defined

1/ Prior to assembly into a cartridge, the primer shall be x-rayed with the side of the cup perpendicular to the rays to determine compliance with the requirements of the applicable drawing. The x-ray negatives, properly identified, shall be forwarded to the procuring activity.

4.3.2.1.3 Carton, Packing, Ammunition for Cartridge, Impulse, M43, Drawing F8797443.

CRITICAL

None defined

MAJOR

None defined

MINOR

201.	Quantity of contents incorrect	Visual
202.	Contents loose	Manual/Visual
203.	Separator or filler missing	Visual
204.	Heat sealer incomplete, loose or distorted	Visual
205.	Cut or damage through impregnated box	Visual
206.	Marking missing, incorrect or illegible	Visual

s/s

Source: https://assist.dla.mil -- Downloaded: 202640-22T07:23Z
Check the source to verify that this is the current version before use.

4.3.2.1.4 Box, Packing, Ammunition for Cartridge, Impulse, M43, Drawing F8797442.

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual/Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

4.3.2.1.5 Carton, Packing, Alternative, Ammunition for Cartridge, Impulse, M43 Drawing C9225902.

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual
202. Carton broken or damaged	Visual
203. Number of units contained incorrect	Visual
204. Marking missing, incorrect, or illegible	Visual
205. Tape incomplete	Visual

4.3.3 Inspection testing.-The following tests shall be performed in accordance with the provisions of 4.4.

Source: https://assist.dla.mil -- Downloaded: 202610-22T07:23Z
Check the source to verify that this is the current version before use.

4.3.3.1 Protective finish.-Unless otherwise specified, sampling plans and procedures in determination of the acceptability of coated parts submitted by a supplier shall be in accordance with provisions set forth in Tables II and III and MIL-C-60539.

4.3.3.2 Airtightness.-Each cartridge shall be subjected to the minimum pressure differential. Failure to comply with the requirements of the drawings shall be cause for rejection of the cartridge.

4.3.3.3 Vibration test - Initial production sample.-Twelve of the 30 assembled x-rayed cartridges of the initial production sample, properly identified, shall be subjected to vibration as directed in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/ of 4.3.2.1.1. Evidence of any disintegration of the primer or propellant or derangement of any component shall be cause for rejection of the lot. When it becomes necessary to make a change in the lot of primers, propellant or igniter after production is initiated, 12 cartridges from the first production lot thereafter shall be tested in accordance with 4.3.3.3.

4.3.3.4 Hardness.-Ten (10) units from each heat-treated lot of components shall be tested. Failure of any unit to comply with the requirements of drawings shall be cause for rejection of the lot of components.

4.3.3.5 Sealant application.-Five (5) cartridges from each production lot shall be selected for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for lot rejection.

4.3.3.6 Functional.-The following tests shall be run by the Government:

4.3.3.6.1 Ballistics.-Thirty (30) units from the initial production sample, and thirty (30) units from subsequent production lots, shall be grouped and temperature conditioned as prescribed in Table IV and 4.4.6.2 and then tested in accordance with 4.4.6.

TABLE IV

Temperature	Initial Production Sample	Production Lot Sample
-65°F to -70°F	10	10
65°F to 75°F	10	10
160°F to 165°F	10	10

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:23Z
Check the source to verify that this is the current version before use.

95

4.3.4 Inspection equipment.-The examination and tests shall be made using equipment listed on IEL-8593275.

4.4 Test methods and procedures.

4.4.1 Protective finish.-Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Airtightness.-The method and procedure for determining airtightness shall be made by use of equipment listed on IEL-8593275.

4.4.3 Vibration.-The method and procedure for performing the vibration test shall be in accordance with Specification MIL-E-5272, Procedure XII.

4.4.4 Hardness.-A hardness testing machine shall be used for determining the hardness values specified on the applicable drawing.

4.4.5 Sealant application.-The cartridge shall be disassembled in such a manner that determination can be made that the sealant has been applied for the field length of the joining surfaces at the junction of the cartridge head and case sidewall and that the propellant cavity is free of sealant.

4.4.6 Functional tests.

4.4.6.1 Ballistics.-Thirty (30) units shall be tested to determine compliance with 3.4.1, 3.4.2, and 3.4.3. The units shall be temperature conditioned as provided in Table IV and 4.4.6.2. Failure of any unit to comply with the requirements of 3.4.1, 3.4.2, and 3.4.3 shall be cause for rejection of the lot.

4.4.6.2 Temperature conditioning.-Cartridge will be loaded into acceptable M2A2 Thrusters with end caps to cover the gas port during the conditioning period. The temperature of the propellant within the cartridge will be considered as the conditioning temperature and will be determined by means of a thermocouple attached to the center of the propellant charge of a specimen cartridge assembled in a test device similar to the thrusters undergoing tests. The conditioning time will be a minimum of one hour longer than the time necessary to bring the elements contained in the specimen cartridge to the required temperature. Upon completion of the required temperature conditioning, the sample will be ballistically tested within 5 minutes after removal from the conditioning chamber. The sample will be reconditioned if this time is exceeded.

95

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:23Z
Check the source to verify that this is the current version before use.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.-Packing and marking shall be as specified by the procuring activity. Level A - Drawing F8797442 and F8797443, Level B - Drawing F8797443 and C9225902.

6. NOTES

6.1 Load.-The expected performance of the cartridge when assembled and tested in the M2A2 Thruster under load conditions simulating those existing in aircraft shall be referred to as load condition in this specification.

6.2 Ordering data.-Procurement documents should specify the title, number and date of this specification.

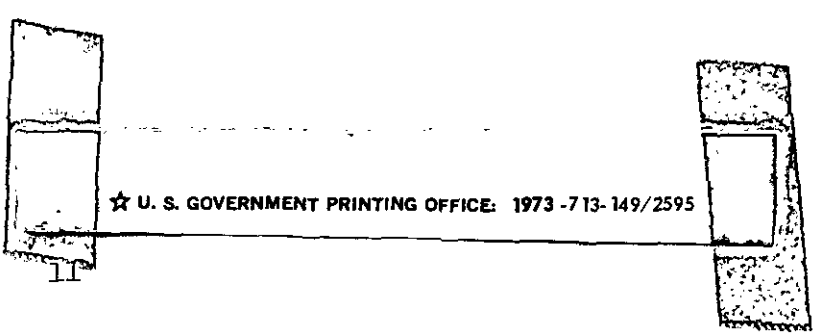
Custodian:

Army - MU

Preparing activity:

Army - MU(FA)

Project NO. 1377-A564



Source: https://assist.dla.mil -- Downloaded: 2026-10-22T07:23Z
Check the source to verify that this is the current version before use.